

Report X

Pade approximation of Gauss function

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1. Introduction

Pade approximation is a numerical method used to approximate a known smooth function by a rational function. Instead of using only a polynomial, as in the case of Taylor approximation, the Pade approximation is expressed as a quotient of two polynomials:

$$R_{N,M}(x) = \frac{P_N(x)}{Q_M(x)}$$

where:

$$P_N(x) = \sum_{i=0}^N a_i x^i$$

and

$$Q_M(x) = \sum_{i=0}^M b_i x^i$$

with the normalization condition:

$$b_0 = 1$$

The main advantage of this approach is that rational functions may often approximate complicated functions better than ordinary polynomial expansions. This is especially useful when the approximated function has a shape which is difficult to reproduce using a single polynomial of limited degree.

In this project, the approximated function was the Gauss function:

$$f(x) = e^{-x^2}$$

The approximation was constructed around $x = 0$, using the Maclaurin series of the Gauss function:

$$e^{-x^2} = \sum_{p=0}^{\infty} \frac{(-1)^p x^{2p}}{p!}$$

From this expansion, the coefficients c_k are equal to:

$$c_k = \begin{cases} \frac{(-1)^{k/2}}{(k/2)!}, & k \text{ even} \\ 0, & k \text{ odd} \end{cases}$$

The coefficients of the rational approximation are obtained by comparing the derivatives of $R_{N,M}(x)$ and $f(x)$ at $x = 0$. This leads to a system of linear equations for the coefficients of the denominator polynomial $Q_M(x)$. After solving this system, the numerator coefficients a_i are computed from the relation:

$$a_i = \sum_{j=0}^i c_{i-j} b_j$$

The goal of this exercise was to compute and analyze the Pade approximation of the Gauss function for several combinations of polynomial degrees.

2. Description of the Numerical Task

The numerical task was to compute the Pade approximation of the function:

$$f(x) = e^{-x^2}$$

on the interval:

$$x \in [-5, 5]$$

The following rational approximations were analyzed:

$$R_{2,2}(x), R_{4,4}(x), R_{6,6}(x), R_{2,4}(x), R_{2,6}(x), R_{2,8}(x)$$

For each pair of indices (N, M) , the coefficients of the Maclaurin series were first generated. Then, a system of linear equations was constructed and solved in order to obtain the coefficients b_i of the denominator polynomial. Finally, the coefficients a_i of the numerator polynomial were calculated.

The computed rational approximation was compared with the exact Gauss function on a dense grid of points in the interval $[-5, 5]$. The results were saved to text files and used to prepare plots.

The main aim was to observe how the degrees of the numerator and denominator influence the quality of approximation. Special attention was paid to the difference between the cases $N = M$ and the cases where the denominator degree is larger than the numerator degree.

3. Numerical Method

The Pade approximation has the form:

$$R_{N,M}(x) = \frac{P_N(x)}{Q_M(x)}$$

where:

$$P_N(x) = a_0 + a_1x + a_2x^2 + \dots + a_Nx^N$$

and

$$Q_M(x) = 1 + b_1x + b_2x^2 + \dots + b_Mx^M$$

The first step of the method is the computation of Maclaurin coefficients c_k for the Gauss function. Since:

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \dots$$

only even powers of x appear in the expansion. Therefore:

$$\begin{aligned} c_0 &= 1 \\ c_1 &= 0 \\ c_2 &= -1 \\ c_3 &= 0 \end{aligned}$$

and so on...

The denominator coefficients were obtained by solving the following system of linear equations:

$$A \cdot x = y$$

where the matrix elements are defined as:

$$A_{i,j} = c_{N-M+i+j+1}$$

and the right-hand side vector is:

$$y_i = -c_{N+1+i}$$

The solution of this system gives the coefficients of the denominator polynomial:

$$\begin{aligned} b_0 &= 1 \\ b_{M-i} &= x_i \end{aligned}$$

After obtaining the vector b , the numerator coefficients were calculated using:

$$a_i = \sum_{j=0}^i c_{i-j} b_j$$

This produces the complete rational approximation $R_{N,M}(x)$, which can then be evaluated for any selected value of x .

4. Implementation

The program was written in C++ and followed the algorithm given in the instruction file. The LAPACK routine LAPACKE_dsysv() was used to solve the symmetric system of linear equations, as suggested in the instruction file.

Main steps of the implementation:

1. Maclaurin coefficients

- coefficients c_k were calculated from the expansion of the Gauss function,
- only even coefficients are different from zero,
- factorial function was used to compute values of $(k/2)!$

2. Matrix construction

- matrix A was created according to the formula from the instruction file,
- right-hand side vector was calculated,
- matrix was stored in column-major order.

3. Solving the system of equations

- system of linear equations was solved in the form:

$$A \cdot x = y$$

- LAPACK routine was used:

$$\text{LAPACKE_dsysv}()$$

- this routine was chosen because the matrix is symmetric.

4. Denominator coefficients

- solution of the system was used to determine coefficients of polynomial $Q_M(x)$,
- vector b was created as:

$$b = [b_0, b_1, \dots, b_M]$$

- first coefficient was set to:

$$b_0 = 1$$

5. Numerator coefficients

- coefficients of polynomial $P_N(x)$ were calculated,
- vector a was created as:

$$a = [a_0, a_1, \dots, a_N]$$

6. Function evaluation

- exact function was calculated as:

$$f(x) = e^{-x^2}$$

- rational approximation was calculated as in the formula on the first page

7. Data export

- data was saved to text files for plotting,
- each file contained:

$$x, f(x), R_{N,M}(x)$$

5. Results

The following cases were analyzed:

- $R_{2,2}(x)$
- $R_{4,4}(x)$
- $R_{6,6}(x)$
- $R_{2,4}(x)$
- $R_{2,6}(x)$
- $R_{2,8}(x)$

Case 1: $R_{2,2}(x)$

- good approximation near $x = 0$,
- visible differences for larger $|x|$,
- approximation does not decrease correctly to zero.

Case 2: $R_{4,4}(x)$

- better accuracy near the center,
- approximation follows the function well close to $x = 0$,
- incorrect behavior appears far from the origin.

Case 3: $R_{6,6}(x)$

- best local accuracy for cases with $N = M$,
- very good agreement near $x = 0$,
- approximation tends to a constant value for large $|x|$.

For $x \rightarrow \pm\infty$:

$$R_{6,6}(x) \rightarrow \frac{a_6}{b_6}$$

Case 4: $R_{2,4}(x)$

- better behavior than cases with $N = M$,
- denominator degree is larger than numerator degree,
- approximation tends to zero for large $|x|$.

Case 5: $R_{2,6}(x)$

- good approximation near the origin,
- better behavior in the tails,
- approximation follows the shape of the Gauss function well.

Case 6: $R_{2,8}(x)$

- best result among cases with $N = 2$,
- good decay for large $|x|$,
- very good qualitative agreement with the exact function.

General observations

- all approximations are accurate near $x = 0$,
- increasing degree improves local accuracy,
- cases with $N = M$ do not reproduce the correct asymptotic behavior,
- cases with $M > N$ approximate the function nicely.

Figure 1. Pade approximation $R_{2,2}(x)$ of Gauss function.

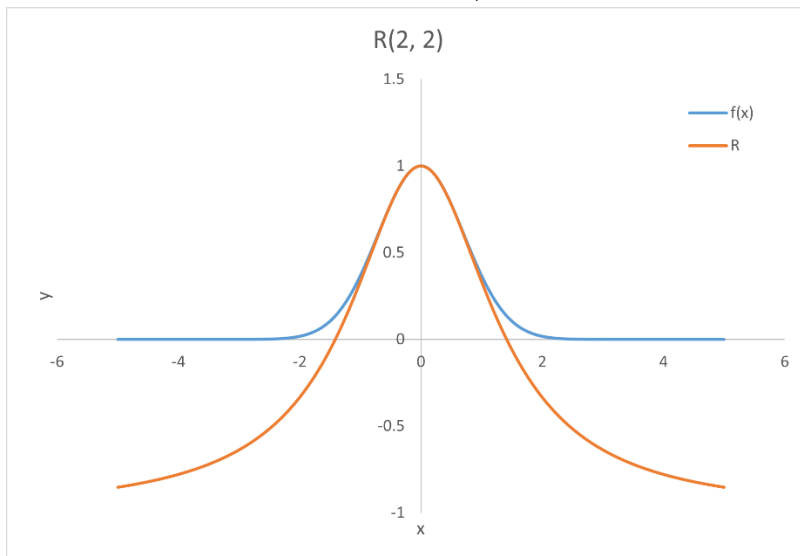


Figure 2. Pade approximation $R_{4,4}(x)$ of Gauss function.

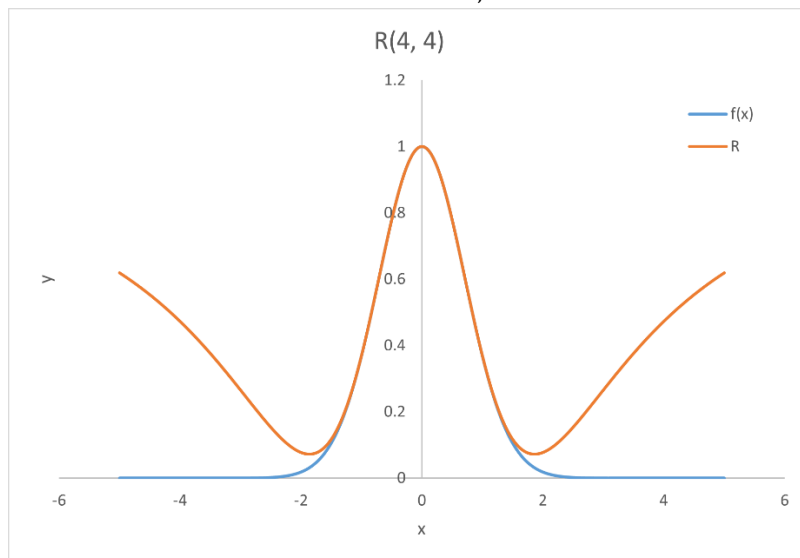


Figure 3. Pade approximation $R_{6,6}(x)$ of Gauss function.

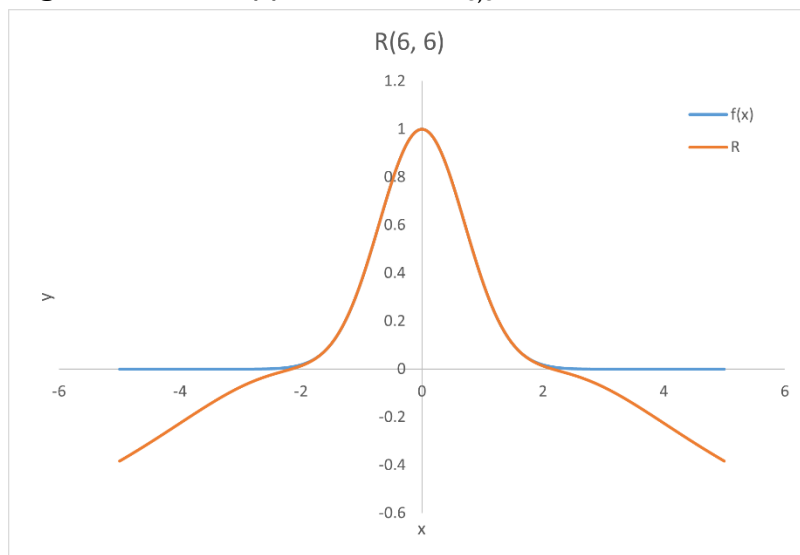


Figure 4. Pade approximation $R_{2,4}(x)$ of Gauss function.

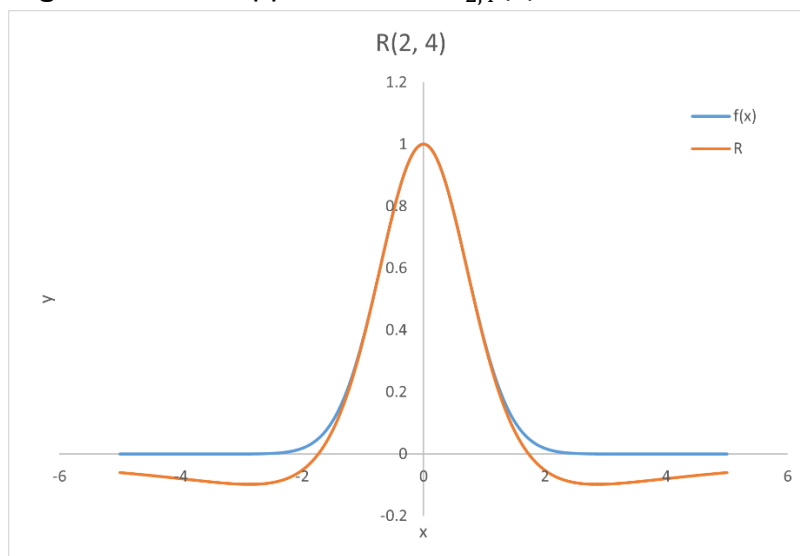


Figure 5. Pade approximation $R_{2,6}(x)$ of Gauss function.

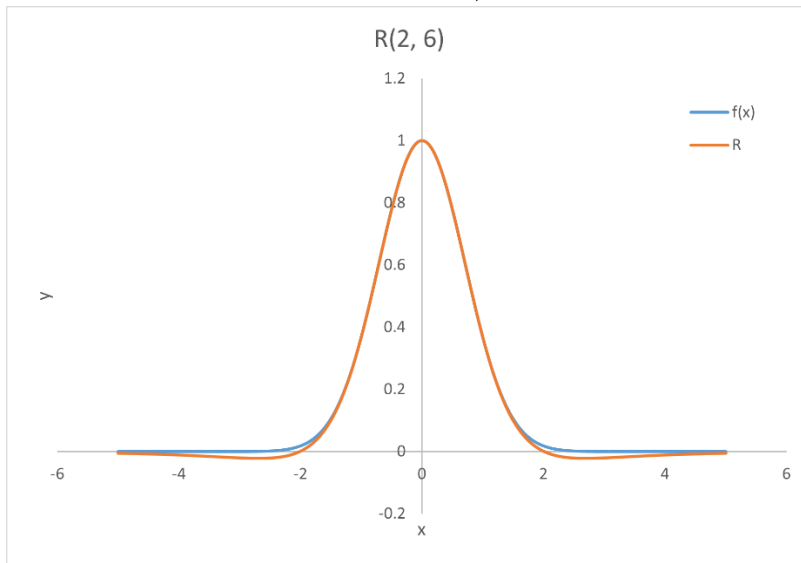
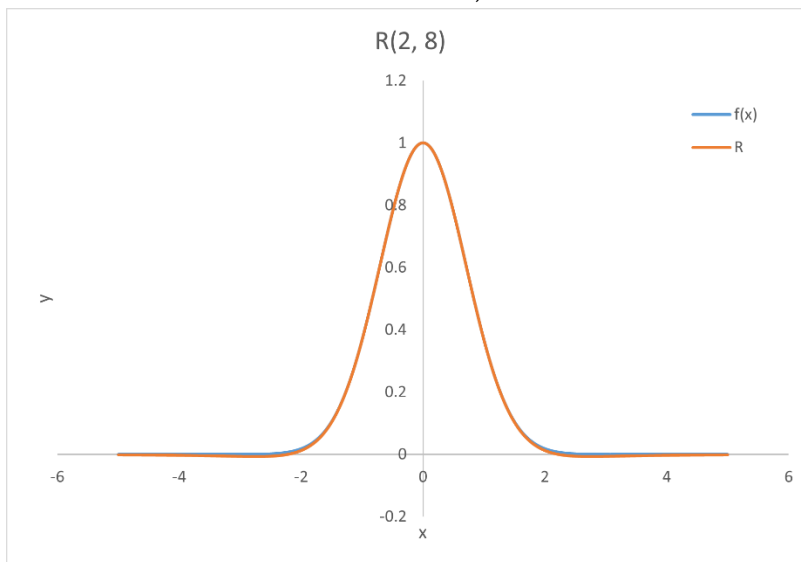


Figure 6. Pade approximation $R_{2,8}(x)$ of Gauss function.



6. Conclusions

Pade approximation provides a good method for local approximation of smooth functions.

The following observations were made:

- approximation is very accurate near $x = 0$,
- increasing degree improves the approximation near the origin,
- cases with $N = M$ do not approximate the Gauss function well globally,
- $R_{6,6}(x)$ tends asymptotically to some constant value,

- exact Gauss function tends to zero for large $|x|$,
- cases with $M > N$ give better behavior far from the origin,
- $R_{2,8}(x)$ gives the best global behavior among tested cases.

The rational function $R_{N,M}(x)$ cannot approximate the Gauss function well for $N = M$, because numerator and denominator have the same degree. Therefore, for large values of x , the approximation tends to a constant value instead of zero.

Overall, Pade approximation works quite well near the expansion point, but the choice of polynomial degrees is important. For the Gauss function, better results are obtained when the denominator degree is larger than the numerator degree.